arXiv:2609.05654v1 [stat.ME] 4 Sep 2026

Robust Community Detection for Noisy Networks with Covariates: Application to Functional Brain Networks

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September 9, 2026

Abstract

Community detection is fundamental to understanding the modular organization in functional brain networks, yet noise in neuroimaging-derived networks and auxiliary node-level covariates pose critical challenges. Existing methods typically either assume networks are noise-free or ignore covariate information. We propose a Bayesian framework for recovering a shared latent community structure from multiple noisy network realizations and auxiliary covariates. The model combines a degree-corrected stochastic block model for the latent network, a block-structured noise model linking noisy observations to latent edges, and a covariate cluster model for node-level attributes. This specification allows anatomical or functional attributes of regions of interest to contribute information when network signals are weak or sparse. We develop an efficient Markov chain Monte Carlo algorithm for posterior sampling and select the number of communities using the widely applicable information criterion, avoiding prior specification of this quantity. Simulation studies demonstrate improved community recovery relative to existing methods across varying noise levels, covariate signal strengths, and numbers of noisy networks, with larger gains when network noise is moderate to high or only a small number of noisy networks is available. Applications to functional brain networks from the Alzheimer's Disease Neuroimaging Initiative and the Human Connectome Project identify biologically interpretable structures and capture disease-related reorganization and individual-level variation.

Key words. Bayesian methods; degree correction; latent graph; MCMC algorithm; network noise; stochastic block model

1 Introduction

The human brain is a highly complex system with organized structural and functional architectures (Bullmore and Sporns, 2009; Rubinov and Sporns, 2010; Sporns, 2013, 2016). Recent advances in magnetic resonance imaging (MRI), particularly functional MRI (fMRI), have enabled in vivo characterization of large-scale brain organization and connectivity (Biswal et al., 1995; Hagmann et al., 2008; Smith et al., 2013; Sporns et al., 2005). A fundamental property of brain organization is its modular structure, where groups of regions of interest (ROIs) form communities supporting specialized functions and interactions (Meunier et al., 2009, 2010). Functional brain network analysis has emerged as a central paradigm for quantifying and interpreting this modular organization (Power et al., 2011; Yeo et al., 2011). This approach has been widely used to characterize individual variation in brain organization (Gordon et al., 2017; Mueller et al., 2013) and changes in functional network organization associated with neurodegenerative diseases such as Alzheimer's disease (AD) (Xu et al., 2024; Seeley et al., 2009; Supekar et al., 2008).

To characterize modular organization, community detection plays a central role in functional brain network analysis. In this framework, the brain is represented as a network, where nodes correspond to ROIs and edges encode the statistical dependencies between regional signals (Logothetis, 2002; Ogawa et al., 1990). Identifying communities provides a data-driven characterization of large-scale brain systems, capturing functionally specialized yet interacting subnetworks. A wide range of community detection methods have been developed, including modularity-based methods (Blondel et al., 2008; Newman, 2006), spectral clustering (Ng et al., 2002; von Luxburg, 2007), and stochastic block models and their extensions (Airoldi et al., 2008; Holland et al., 1983).

Despite the widespread use of these community detection methods, several challenges remain. At the individual level, functional connectivity is not directly observed but estimated from neuroimaging data, introducing both measurement and estimation uncertainty (Murphy et al., 2013; Power et al., 2012; Smith et al., 2011). Consequently, repeated measurements from the same individual can yield noisy network realizations, and failure to account for such noise may reduce the accuracy of community detection. At the population level, a central goal is to recover a shared organizational structure from a collection of individual networks, which can be viewed as noisy realizations of an underlying population level template. This problem is particularly relevant in AD studies, where comparing community structure across disease stages provides insights for characterizing progressive network disruption and identifying deviations from normative functional organization (Xu et al., 2024; Sheng et al., 2024).

In addition, neuroimaging studies often collect auxiliary ROI-level covariates, including spatial, anatomical, and functional attributes of brain regions (Binkiewicz et al., 2017; Glasser et al., 2016; Schaefer et al., 2018). Such information complements network topology and can improve community recovery, particularly when network signals are sparse or weak (Deshpande et al., 2018; Yan and Sarkar, 2021). Integrating these covariates with network data, while simultaneously accounting for network noise at both the individual and population levels, remains a critical challenge.

Existing approaches address these challenges only partially. Classical community detection methods rely solely on network topology and assume the observed network is noise free. Covariate-assisted approaches (e.g., Binkiewicz et al., 2017; Hu and Wang, 2024; Louit et al.,

2025; Shen et al., 2025) use node-level covariates but do not account for network noise. Recent methods that account for network noise (e.g., Hu et al., 2024) do not incorporate node covariate information. To the best of our knowledge, no existing community detection framework simultaneously accounts for network noise and leverages node-level covariate information.

Our contribution in this paper is to provide a Bayesian community detection framework that jointly accounts for network noise and leverages nodal covariate information, without requiring prior knowledge of the number of communities. The method treats the true network as latent and has three key components. We employ a degree-corrected stochastic block model (DC-SBM, Peng and Carvalho, 2016) to characterize community structure in the underlying network. The DC-SBM is integrated with a block-structured network noise model (Le et al., 2018) that relies on noisy networks to inform latent communities. These components are combined with a covariate cluster model, adapted from existing approaches (Jin and Wang, 2016; Jin et al., 2017; Shahbaba and Neal, 2009), to incorporate node-level covariate information for community detection. This joint specification uses the complementary information in noisy networks and node covariates to improve estimation of latent community structure. For brevity, we refer to the proposed approach as RCDC, Robust Community Detection for noisy networks with Covariates. We evaluate RCDC through simulation studies with varying network noise levels and numbers of noisy networks, and through applications to functional brain networks from Alzheimer's Disease Neuroimaging Initiative (ADNI) study (Mueller et al., 2005) and Human Connectome Project (HCP, Van Essen et al., 2013).

The organization of this paper is as follows. In Section 2 we present the proposed methodology and the Markov chain Monte Carlo (MCMC) algorithm. Section 3 reports the practical performance of our method through simulation studies. In Section 4 we apply our method to functional brain networks from the ADNI and HCP studies. Finally, we conclude in Section 5 with a discussion of future directions for this work. The details of the computation framework are relegated to Web Appendix A.

2 Methodology

We present our proposed model first and then the associated computation framework.

2.1 Model Specification

Let $G = (V, E)$ denote the true underlying undirected graph on n nodes, with vertices $V = \{1, \ldots, n\}$ and a set E of edges, where elements of E are unordered pairs $\{i, j\}$ of distinct vertices $i,\ j \in V$. Suppose we have data of M noisy graphs, denoted by $G^{(m)} = (V, E^{(m)})$ for $m = 1, \ldots, M$, where we implicitly assume that the vertex set V is known. Denote the adjacency matrix of G by $\boldsymbol{A} = (A_{ij})_{n\times n}$ and that of $G^{(m)}$ by $\tilde{\boldsymbol{A}}^{(m)} = (\tilde{A}_{ij}^{(m)})_{n\times n}$. Hence $A_{ij} = 1$ if there is a true edge between the i-th vertex and the j-th vertex, and 0 otherwise, while $\tilde{A}_{ij}^{(m)} = 1$ if there is an edge in the noisy graph $G^{(m)}$ between the i-th vertex and the j-th vertex, and 0 otherwise. We assume throughout that G and $G^{(m)}$, $m = 1, \ldots, M$, are simple, i.e., that they possess neither multi-edges nor self-loops. In addition, assume that we have covariate information $\boldsymbol{x}_i \in \mathbb{R}^p$ for each node $i = 1, \ldots, n$. Our goal is to detect

communities by leveraging this nodal covariate information while accounting for noise in $G^{(m)}$, $m = 1, \ldots, M$.

We model the structure in $\boldsymbol{A}$ through one of the most commonly used network community models, DC-SBM. Let K be the number of communities. Note that our method does not require prior knowledge of K, and the criterion for selecting K is described in Section 2.2. Define a symmetric matrix $\boldsymbol{B} = (B_{kl})_{K\times K}$, a node membership vector $\boldsymbol{z} = (z_1, \ldots, z_n) \in [K]^n$ where $[K] = \{1, \ldots, K\}$ for any $K \in \mathbb{N}$, and a vector of node degree-correction parameters $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_n)^\top$. For all $1 \le i < j \le n$, we assume

$$A_{ij} \mid \boldsymbol{z}, \boldsymbol{B}, \boldsymbol{\theta} \sim \text{Bern}\left(\text{logit}^{-1}\left(B_{z_i z_j} + \theta_i + \theta_j\right)\right), \tag{1}$$

where $\text{Bern}(\cdot)$ denotes the Bernoulli distribution. On the logit scale, $\boldsymbol{B}$ captures within and between community connectivity. Following Peng and Carvalho (2016), we impose an identifiability constraint by setting $B_{11} = \cdots = B_{KK} = 0$, yielding

$$A_{ij} \mid \boldsymbol{z}, \boldsymbol{B}, \boldsymbol{\theta} \sim \text{Bern}\left(\text{logit}^{-1}\left\{B_{z_i z_j}\mathbb{I}\left(z_i \neq z_j\right) + \theta_i + \theta_j\right\}\right), \tag{2}$$

where $\mathbb{I}(\cdot)$ denotes the indicator function. We assign the following prior to $\boldsymbol{\theta}$ and $\{B_{kl} : 1 \le k < l \le K\}$:

$$\left(\{B_{kl} : 1 \le k < l \le K\}, \boldsymbol{\theta}^\top\right)^\top \sim \mathbb{I}\left(\{B_{kl} : 1 \le k < l \le K\} \le 0\right) \cdot \mathcal{N}(0, \sigma^2 \boldsymbol{I}_{n+K(K-1)/2}), \tag{3}$$

where $\mathcal{N}(\cdot)$ denotes the Gaussian distribution and $\boldsymbol{I}$ denotes the identity matrix. The constraint $B_{kl} \le 0$ encourages edge densities between communities to be no greater than those within communities, which is essential for community detection.

Adapting the block-structured network noise assumption of Le et al. (2018), we specify the model for $\tilde{\boldsymbol{A}}^{(m)}$, $m = 1, \ldots, M$, as follows: for $1 \le i < j \le n$,

$$\begin{aligned} \Pr\left(\tilde{A}_{ij}^{(m)} = 1 \mid A_{ij} = 0,\ z_i = k,\ z_j = l\right) &= \alpha_{kl}, \\ \Pr\left(\tilde{A}_{ij}^{(m)} = 0 \mid A_{ij} = 1,\ z_i = k,\ z_j = l\right) &= \beta_{kl}. \end{aligned} \tag{4}$$

Here α_{kl} and β_{kl} represent the false positive and false negative rates, respectively, for node pairs whose endpoints belong to communities k and l. We assume symmetry, i.e., $\alpha_{kl} = \alpha_{lk}$ and $\beta_{kl} = \beta_{lk}$, for all $k \neq l, k, l = 1, \ldots, K$. By convention, we assign the beta distributed priors to the error rates:

$$\alpha_{kl} \sim \text{Beta}(a_\alpha, b_\alpha),\ \beta_{kl} \sim \text{Beta}(a_\beta, b_\beta),\ 1 \le k \le l \le K, \tag{5}$$

where $\text{Beta}(\cdot)$ denotes the Beta distribution.

We model the covariates $\boldsymbol{x}_i$ using a standard cluster model (Jin and Wang, 2016; Jin et al., 2017), and adapt the conditional independence assumption in Hu and Wang (2024). Specifically, the covariates $\boldsymbol{x}_i$ are independently distributed as

$$\boldsymbol{x}_i \mid \boldsymbol{z} \sim F_{z_i}, \tag{6}$$

where F_k denotes a distribution for community k, $k \in [K]$. When p is moderate to large, estimation of the joint distribution is both statistically and computationally challenging. We adapt the approach of Shahbaba and Neal (2009) to address this. Specifically, we assume the covariates are independent within components, i.e., $p(\boldsymbol{x}_i|z_i) = \prod_{r=1}^{p} p(x_{ir}|z_i)$, where x_{ir} denotes the rth covariate of node i. We provide details below for normal and categorical distributions, which are commonly used. The model can also be extended to accommodate other distributions. For the normal distribution case, we specify

$$\begin{aligned} x_{ir}^{\text{cont}} \mid z_i = k, \mu_{kr}, \tau_{kr}^2 &\sim \mathcal{N}(\mu_{kr}, \tau_{kr}^2), \\ \mu_{kr} \mid \tau_{kr}^2 &\sim \mathcal{N}(\mu_{0r},\ \iota_{0r}\tau_{kr}^2), \\ \tau_{kr}^2 &\sim \mathcal{IG}(a_{\tau r}, b_{\tau r}), \end{aligned} \tag{7}$$

where x_{ir}^{cont} denote a continuous covariate of node i, and $\mathcal{IG}(\cdot)$ denotes the inverse gamma distribution. Let x_{is}^{cat} be a categorical covariate of node i and $x_{is}^{\text{cat}} \in \{1, \ldots, C_s\}$. Denote the corresponding vector of category probabilities in community k by $\boldsymbol{\xi}_{ks} = (\xi_{ks1}, \ldots, \xi_{ksC_s})$, i.e.,

$$\Pr\left(x_{is}^{\text{cat}} = c \mid z_i = k, \boldsymbol{\xi}_{ks}\right) = \xi_{ksc},\ c = 1, \ldots, C_s. \tag{8}$$

We assign the following prior to $\boldsymbol{\xi}_{ks}$:

$$\boldsymbol{\xi}_{ks} \sim \text{Dir}(\gamma_s \mathbf{1}_{C_s}), \tag{9}$$

where $\text{Dir}(\cdot)$ denotes the Dirichlet distribution and $\mathbf{1}$ denotes the all-ones vector.

Following Peng and Carvalho (2016), we assign a constrained multinomial distributed prior to $\boldsymbol{z}$:

$$p(\boldsymbol{z} \mid \boldsymbol{\pi}) \propto \prod_{k=1}^{K} \mathbb{I}(N_k > 1) \prod_{i=1}^{n} \pi_k^{\mathbb{I}(z_i = k)}, \tag{10}$$

where $\boldsymbol{\pi} = (\pi_1, \ldots, \pi_K)$ is a vector of prior probabilities over K labels, and $N_k = \sum_{i=1}^{n} \mathbb{I}(z_i = k)$ denotes the size of community k. The constraint $N_k > 1$ is imposed to ensure model identifiability. In addition, we assign a Dirichlet distributed prior to $\boldsymbol{\pi}$:

$$\boldsymbol{\pi} \sim \text{Dir}\left(\frac{d}{K}\mathbf{1}_K\right). \tag{11}$$

2.2 Computation

We develop an efficient MCMC algorithm for sampling the parameters in our model. To facilitate the sampling, we augment Pólya-Gamma latent variables for Bernoulli distributed variables A_{ij}, $1 \leq i < j \leq n$. All parameters are sampled using the Gibbs sampler. We adopt the label remapping approach in Peng and Carvalho (2016) to address the label identifiability issue, and motivated by Le et al. (2018), propose an initialization strategy to accelerate MCMC convergence. Details of the computation framework can be found in the Web Appendix A.

Following similar works (Merkle et al., 2019; Tan et al., 2024; Louit et al., 2025), we choose the number of communities K by the widely applicable information criterion (WAIC, Watanabe and Opper, 2010),

$$\begin{aligned}\text{WAIC} = &-2\sum_{m=1}^{M}\sum_{1\le i<j\le n}\left\{\log\Big(\mathbb{E}_{\text{post}}\big[p(\tilde{A}_{ij}^{(m)} \mid \Theta)\big]\Big) - \text{Var}_{\text{post}}\big[\log p(\tilde{A}_{ij}^{(m)} \mid \Theta)\big]\right\} \\ &-2\sum_{i=1}^{n}\left\{\log\Big(\mathbb{E}_{\text{post}}\big[p(\boldsymbol{x}_i \mid \Theta)\big]\Big) - \text{Var}_{\text{post}}\big[\log p(\boldsymbol{x}_i \mid \Theta)\big]\right\}, \end{aligned} \tag{12}$$

where Θ denotes all model parameters, $\mathbb{E}_{\text{post}}$ and Var_{post} denote the expectation and variance under the posterior distribution of Θ, respectively. We choose K by minimizing the WAIC.

We develop an open-source and user-friendly `R` package, `rcdc`, for implementing the proposed method. The package is publicly available on GitHub at https://github.com/zyhu888/rcdc.

3 Simulation Studies

We conducted simulations to evaluate the performance of our model in comparison with several existing methods under a range of settings, varying the number of noisy networks and their noise levels, as well as the strength of the network structure and the signal in the node covariates.

3.1 Data Generation

We generated the true network from the DC-SBM in (2) with $B_{kl} = b$ for $k \neq l$, $k, l = 1, \ldots, K$, and $\theta_i \mid z_i = k \sim \mathcal{N}(\mu_k^\theta, 0.1)$, $i = 1, \ldots, n$. We set $n = 200$ nodes with $K = 8$ equally sized communities. The parameters $(\mu_1^\theta, \ldots, \mu_8^\theta)$ were fixed at $(0.5, 0.3, 0.1, -0.1, -0.3, -0.5, -0.7, -0.9)$. The parameter b controls the magnitude of disparity between within- and between-community connectivities. We considered two levels of the network structure: a strong community structure with $b = -2$ and a weak community structure with $b = -1.7$.

Noisy networks were generated from model (4), with within- and between-community false positive and false negative probabilities given by

$$\alpha_{kl} = \begin{cases} \rho_\alpha, & \text{if } k = l, \\ \rho_\alpha\zeta_\alpha, & \text{if } k \neq l, \end{cases} \qquad \beta_{kl} = \begin{cases} \rho_\beta, & \text{if } k = l, \\ \rho_\beta\zeta_\beta, & \text{if } k \neq l. \end{cases}$$

Here, ρ_α and ρ_β control the overall noise level, and ζ_α and ζ_β capture the relative magnitude of within- versus between-community noise. We considered various values of these parameters to examine the effect of network noise on the performance of our method. In addition, we varied the number of noisy networks to assess its impact.

We considered a mix of continuous and categorical covariates, with a total of $p = 10$ covariates. The continuous covariates $\boldsymbol{x}_i^{\text{cont}} \in \mathbb{R}^8$ were generated from a multivariate normal distribution (MVN): $\boldsymbol{x}_i^{\text{cont}} | z_i = k \sim \text{MVN}(\boldsymbol{\mu}_k, \tau^2 \boldsymbol{I}_8)$, where $\boldsymbol{\mu}_k = (\mu_{k1}, \ldots, \mu_{k8})$. The cluster

centers $\boldsymbol{\mu}_k$ were positioned at eight vertices of a 7-simplex centered at the origin with fixed edge length 10. The parameter τ controls the strength of the continuous covariate signal. We further considered two categorical covariates, where each covariate $x_{is}^{\text{cat}} \in \{1, \ldots, C_s\}$ was generated according to (8) with $C_s = 8$, $\xi_{ksc} = \xi$ if $c = k$, and $\xi_{ksc} = (1 - \xi)/(C_s - 1)$ for all $c \neq k$. The parameter ξ controls the strength of the categorical covariate signal. We considered two levels of covariate signal: a strong case with $\tau = 3$ and $\xi = 0.7$, and a weak case with $\tau = 3.5$ and $\xi = 0.6$.

Clustering accuracy was evaluated using the Adjusted Rand Index (ARI, Rand, 1971), which quantifies the agreement between the estimated and true community assignments. We set fairly uninformative priors: $\sigma = 10$ for $(\boldsymbol{B}, \boldsymbol{\theta})$; $\alpha_{kl}, \beta_{kl} \sim \text{Beta}(1, 1)$; $\mu_{0r} = 0$ and $\iota_{0r} = 1$ for μ_{kr}; $\tau_{kr}^2 \sim \mathcal{IG}(1, 0.01)$; $\gamma_s = 1$ for $\boldsymbol{\xi}_{ks}$; and $d = 1$ for $\boldsymbol{\pi}$. We ran MCMC chains of 1000 samples. We discarded the first 500 samples as burn-in and assessed convergence using trace plots.

3.2 Methods to Compare

Our method was compared with several approaches. We included an existing community detection method for noisy networks, the group-based binary mixture (GBM) modeling approach (Hu et al., 2024). We also considered covariate-assisted community detection methods, namely the covariate-assisted spectral clustering (CASC) algorithm (Binkiewicz et al., 2017), the covariate-assisted spectral clustering on ratios of eigenvectors (CASCORE) (Hu and Wang, 2024), the Bayesian community detection for networks with covariates (BCDC) (Shen et al., 2025), and the covariate-assisted latent factor stochastic block model (CALF-SBM) (Louit et al., 2025). In addition, we included methods using a single data source: k-means algorithms (k-means) (Lloyd, 1982) applied only to covariates, and spectral clustering (SPC) (Ng et al., 2002) applied only to adjacency matrix. Since CASC, CASCORE, BCDC, CALF-SBM, and SPC each take one network as input, we constructed an adjacency matrix $\boldsymbol{A}^{\text{est}}$ with entries $A_{ij}^{\text{est}} = \mathbb{I}(\sum_{m=1}^{M} \tilde{A}_{ij}^{(m)} \geq M/2)$, and used it as their network input. We gave the competing methods that cannot handle unknown K (i.e., CASC, CASCORE, k-means and SPC) an additional advantage by assuming the knowledge of the true number of communities.

All methods have implementations in R (R Core Team, 2026). Specifically, k-means was implemented via the `stats` package, and SPC and CASC were performed using the `c4` package (Hu et al., 2026). We used the `CASCORE` package (Hu and Wang, 2023), the `bcdc` package (Shen et al., 2026), and the `calfsbm` package (Louit et al., 2024) for the corresponding methods. For GBM, we used the authors' publicly available code.

3.3 Results

Figure 1 shows the mean ARI with the interquartile range for RCDC and competing methods, averaged over 50 replicates, under homogeneous network noise (i.e., $\zeta_\alpha = \zeta_\beta = 1$). For clarity of presentation, we omit CALF-SBM, as it consistently performs worse than or comparably to the other covariate-assisted community detection method that accommodates unknown K (i.e., BCDC). We also omit CASC, which performs worse than or comparably to the other covariate-assisted method that requires knowledge of K (i.e., CASCORE) in almost all settings. Due to instability of the expectation–maximization algorithm in GBM,

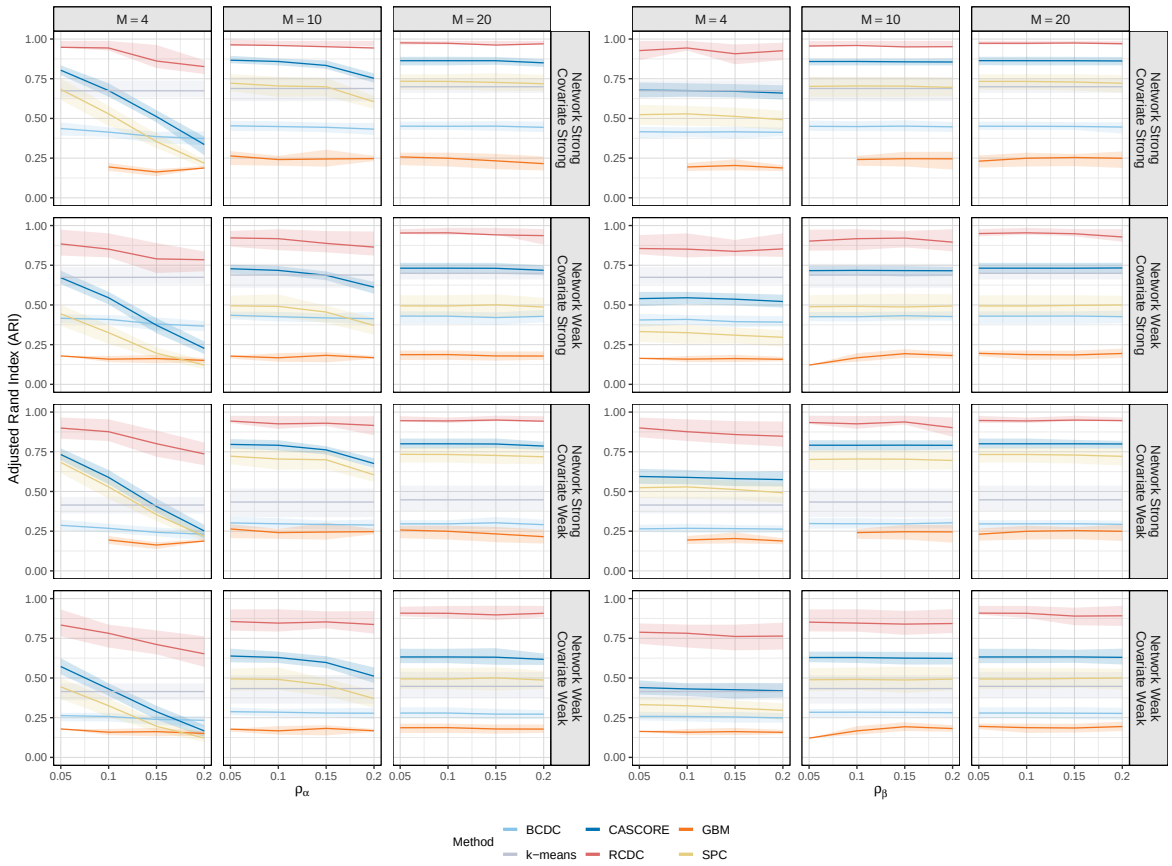


Figure 1: ARI results for different methods under homogeneous network noise ($\zeta_\alpha = \zeta_\beta = 1$). Left panel: fixing false negative rate at 0.1 ($\rho_\beta = 0.1$) and varying the other settings. Right panel: fixing false positive rate at 0.1 ($\rho_\alpha = 0.1$) and varying the other settings.

results could not be obtained for some parameter settings, and settings with fewer than 50 successful replicates are summarized using the mean of the available replicates. In nearly all other settings, GBM has the lowest ARI. RCDC outperforms all competing methods across all settings, including varying network noise levels, numbers of noisy networks, and network structural and covariate signal strengths.

The left and right panels in Figure 1 show the effect of homogeneous false positive and false negative rates when fixing either false positive rate or false negative rate at 0.1 (i.e., ρ_α or $\rho_\beta = 0.1$) and varying the other settings. The ARIs of CASCORE and SPC decrease substantially as the false positive rate increases when the number of noisy networks M is 4 (the left panel, column 1), indicating sensitivity to false positive edges that weaken the community structure. This effect becomes less pronounced as M increases, corresponding to less noisy input networks. When the false negative rate increases (the right panel), the performance of all methods remains relatively stable. One possible explanation is that increasing the false negative rate removes both within- and between-community edges, and the true networks are relatively sparse, thus the strength of the community structure remains largely unchanged. Overall, our method demonstrates robust performance across settings.

The ARIs under heterogeneous network noise are presented in Figure 2. We omit CALF-SBM and CASC again for clarity of presentation, and some GBM results are unavailable due to algorithmic instability. Across all settings, RCDC consistently achieves the highest ARI, including scenarios with varying degrees of network noise heterogeneity, different numbers of noisy networks, and varying strengths of structural and covariate signals.

The left and right panels in Figure 2 show the effects of heterogeneous false positive and false negative rates, respectively. We set the overall noise level parameters $\rho_\alpha = \rho_\beta = 0.1$, while fixing either the false positive or false negative rate at homogeneity (ζ_α or $\zeta_\beta = 1$) and varying the other settings. When $M = 4$, the ARIs of CASCORE and SPC decrease substantially as the ratio of between- to within-community false positive rates ζ_α increases, and increase when the ratio of between- to within-community false negative rates ζ_β increases. This pattern implies that both methods are particularly sensitive to between-community edges, which weaken the underlying community structure. This pattern becomes less pronounced as M increases, consistent with reduced noise in the input networks.

GBM shows a different form of sensitivity under heterogeneous network noise. Although GBM explicitly models network noise, its performance is highly sensitive to both the ratio of between- to within-community noise levels and the number of noisy networks. Specifically, GBM performs poorly when $M = 4$ and displays a non-monotonic pattern when $M = 10$ and 20, with the lowest ARI occurring around $\zeta_\alpha, \zeta_\beta = 1$ (i.e., homogeneous network noise). This suggests that GBM requires a relatively large number of noisy networks and sufficiently strong structural signal to effectively account for network noise. The ARIs of RCDC are relatively similar across all settings, which implies that its performance is robust to heterogeneity in community-structured network noise.

In summary, these results suggest that leveraging both covariates and noisy networks to infer the underlying community structure yields substantial gains compared to existing approaches that either lack noise correction or rely on a single data source.

4 Data Application

We applied the proposed method to analyses of two functional brain network datasets, one from the ADNI study (Mueller et al., 2005) and the other from the HCP (Van Essen et al., 2013).

4.1 Application to ADNI Data

We applied our method to ROI-level functional brain networks with brain region covariate information constructed from data in the ADNI database (`https://adni.loni.usc.edu`). The ADNI is a large, multi-center longitudinal study designed to investigate the progression of AD using neuroimaging, biomarkers, and clinical assessments (Weiner et al., 2013, 2017). We used data from the ADNI Phase 3 (ADNI-3) cohort. Resting-state functional magnetic resonance imaging (rs-fMRI) data were processed using the ConnectomeMapper3 pipeline built in Nipype, and details of packages are available in Tourbier et al. (2022) and related literature. The procedure for obtaining blood–oxygen–level–dependent (BOLD) time signals of ROIs defined by the Lausanne 2018 parcellation is described in Xu et al. (2024). Covariate

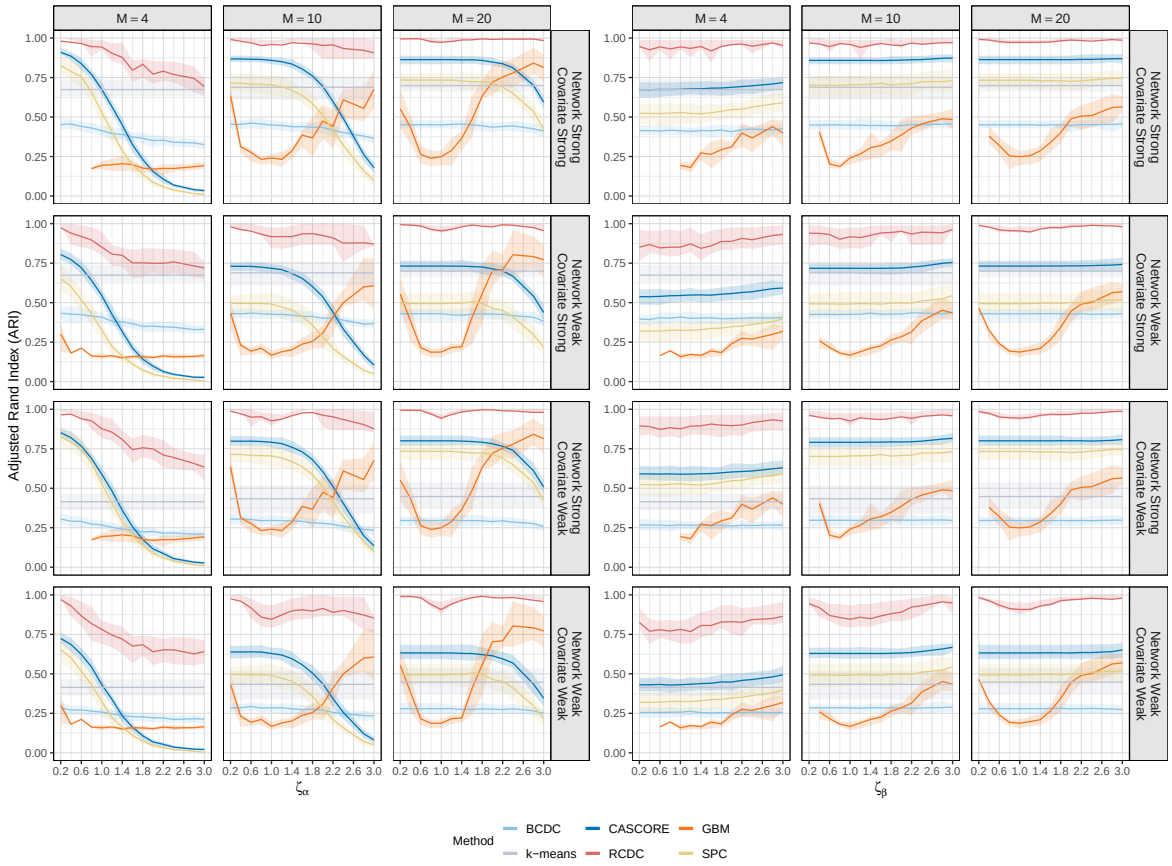


Figure 2: ARI results for different methods under heterogeneous network noise, with overall noise level parameters $\rho_\alpha = \rho_\beta = 0.1$. Left panel: fixing false negative rate at homogeneity ($\zeta_\beta = 1$) and varying the other settings. Right panel: fixing false positive rate at homogeneity ($\zeta_\alpha = 1$) and varying the other settings.

Table 1: Demographic characteristics of subjects in the ADNI dataset by diagnosis group.

Characteristic	HC (n=188)	MCI (n=150)	AD (n=53)
Age, years (mean ± SD)	70.6 ± 6.6	71.7 ± 7.1	74.7 ± 7.0
Gender, n (%)			
Female	98 (52.1)	68 (45.3)	23 (43.4)
Male	90 (47.9)	82 (54.7)	30 (56.6)
Race, n (%)			
White	168 (89.4)	142 (94.7)	51 (96.2)
Black	13 (6.9)	3 (2.0)	0 (0.0)
Asian	1 (0.5)	2 (1.3)	1 (1.9)
Indian/Alaskan	1 (0.5)	0 (0.0)	0 (0.0)
More than one	4 (2.1)	2 (1.3)	1 (1.9)
Unknown	1 (0.5)	1 (0.7)	0 (0.0)
Ethnicity, n (%)			
Hispanic/Latino	11 (5.9)	5 (3.3)	1 (1.9)
Not Hispanic/Latino	174 (92.6)	145 (96.7)	52 (98.1)
Unknown	3 (1.6)	0 (0.0)	0 (0.0)
APOE4 status, n (%)			
Non-carrier	120 (63.8)	85 (56.7)	21 (39.6)
Carrier	66 (35.1)	58 (38.7)	29 (54.7)
Unknown	2 (1.1)	7 (4.7)	3 (5.7)

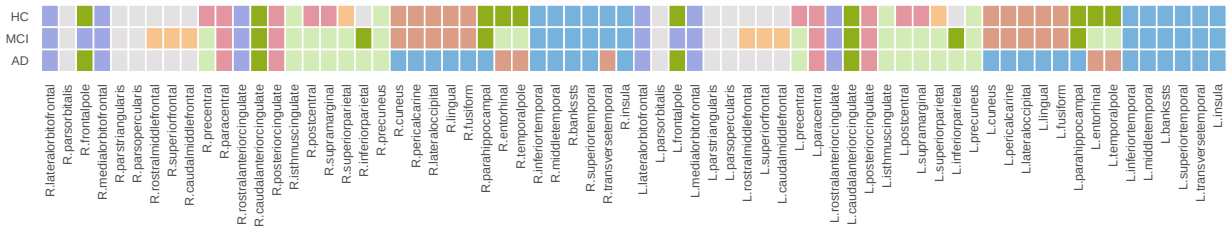
Note: SD denotes standard deviation.

information, including cortical volume, surface area, and average cortical thickness for 68 Desikan–Killiany regions, was obtained from the UC Berkeley processing pipeline (Landau et al., 2015) and matched to imaging scan dates.

For each subject, we constructed a functional brain network for the 68 regions based on the Fisher z-transformed Pearson correlation between BOLD time signals of pairs of ROIs, thresholded at 0.4 (Garrison et al., 2015; Wang et al., 2014). We excluded subjects with missing BOLD time series or diagnostic status. Our analysis included 391 subjects, consisting of 188 healthy control (HC), 150 mild cognitive impairment (MCI), and 53 dementia (AD). The demographic characteristics are summarized in Table 1. We applied our model to detect communities for each diagnostic group.

Clustering results were first compared to 7 functional Resting-State Networks (RSNs) in the Yeo Atlas (Yeo et al., 2011), which were derived from rs-fMRI data collected from healthy young adults. Agreement with RSNs was highest in the HC group (0.340), and decreased progressively in the MCI (0.209) and AD (0.150) groups, indicating increasing disruption of functional community structure with disease progression. To further examine differences in

(a) Community assignments for ROIs (R: right hemisphere; L: left hemisphere).



(b) Spatial depiction of detected communities.

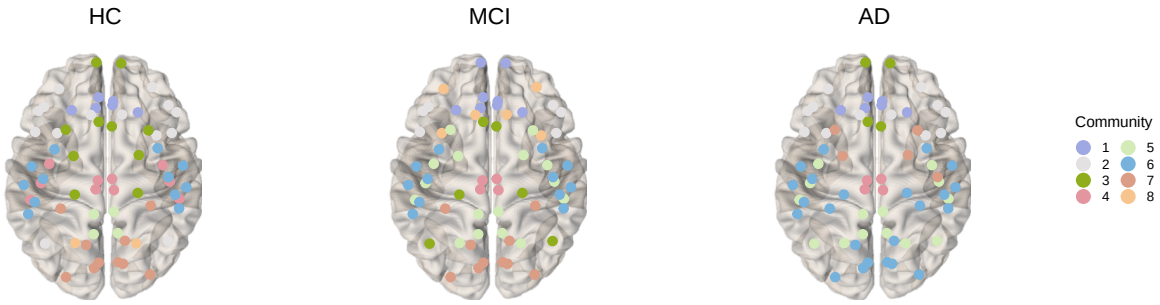


Figure 3: Community structure of functional brain networks in HC, MCI, and AD groups.

community structure across diagnostic groups, we quantified the similarity of MCI and AD to HC using ARI. Similarity also decreased with disease severity, with higher similarity in MCI (0.577) than in AD (0.506), indicating a gradual reorganization of functional community structure from HC to MCI to AD.

We also examined region-specific community assignments to identify localized patterns of reorganization (Figure 3). The number of detected communities was $K = 8$ for HC and MCI and $K = 7$ for AD. Community labels for MCI and AD were aligned with those of HC using the Hungarian algorithm (Kuhn, 1955) to maximize overlap for comparison. Figure 3a shows community assignments for ROIs in both hemispheres for HC, MCI, and AD groups, where each region is color-coded according to its community membership. Figure 3b visualizes the consensus communities in 3D Montreal Neurological Institute space, with nodes overlaid on a brain template to illustrate functional topography across the three diagnostic groups.

Several ROIs known to be vulnerable in AD showed consistent and progressive shifts in community affiliation across disease stages (Figure 3a). In particular, the entorhinal cortex, a key medial temporal structure for memory processing, exhibited clear stage dependent changes in community affiliation (from community 3 to 5, then 7), consistent with its role as an early site of neurodegeneration (Braak and Braak, 1991; Killiany et al., 2000). In addition, the inferior parietal cortex, a central hub of the default mode network, also exhibited consistent transitions across stages (from community 2 to 3, then 5), reflecting disruption of functional networks that are characteristically affected in AD (Buckner et al., 2005; Greicius et al., 2004). In contrast, prefrontal regions, including the rostral middle frontal, superior frontal, and caudal middle frontal cortices, exhibited stage-specific changes, forming a distinct

community in the MCI stage. Such reorganization in prefrontal networks has been reported in MCI and is often interpreted as potential compensatory recruitment in response to early brain disruption (Grady et al., 2003; Qi et al., 2010).

Finally, Figure 3b shows that, in the AD group, community boundaries became markedly less distinct, accompanied by the emergence of a large, widely distributed community (blue) that incorporated many regions assigned to communities 6 and 7 in HC and MCI groups. This pattern indicates reduced network segregation and increased global integration, consistent with widespread disruption of functional organization in advanced stages of AD (Brier et al., 2014).

4.2 Application to HCP Data

We applied our method to ROI-level functional brain networks with node-level covariate information constructed from the HCP data. The HCP is a large-scale project designed to characterize the structural and functional connectivity of the human brain (Van Essen et al., 2013). We used rs-fMRI data from the young adult data release of the HCP, with four runs acquired per subject. BOLD time series were obtained for 360 ROIs defined by the HCP multimodal parcellation atlas (Glasser et al., 2016), following the procedure described in Akiki and Abdallah (2019). We considered ROI-level covariate information, including cortical thickness, cortical curvature, and sulcal depth.

We constructed one functional brain network per subject per run using the same Fisher z-transformed Pearson correlation approach described in Section 4.1. We selected thresholds to yield network densities around 0.15 (Garrison et al., 2015). For each subject, we treated the four functional brain networks as noisy realizations of the underlying network. Our analysis included 376 subjects with complete BOLD time series and covariate information. The demographic characteristics are summarized in Table 2. We applied our method to detect communities for each subject.

Figure 4 displays the ARI heat map. Off-diagonal entries show pairwise ARIs between subjects, and diagonal entries show the ARI between each subject-specific community assignment and Cole–Anticevic Brain-wide Network Partition (CAB-NP, Ji et al., 2019), a partition of the ROIs into 12 canonical functional networks. The subject-specific communities showed moderate agreement with CAB-NP, and the pairwise ARIs were positive and varied across subjects. These implied that our method detects community structures that capture shared functional organization and preserve subject-level heterogeneity. Females showed higher ARI than males for CAB-NP, and pairwise ARIs were higher among female subjects than among male subjects. These findings are consistent with previous studies reporting differences between females and males in functional brain networks (Filippi et al., 2013; Zhang et al., 2018).

5 Discussion

We have developed a Bayesian framework for detecting the underlying community structure from noisy networks with node-level covariates. By jointly leveraging information from noisy graphs and node covariates, we provide a robust approach to recovering the community

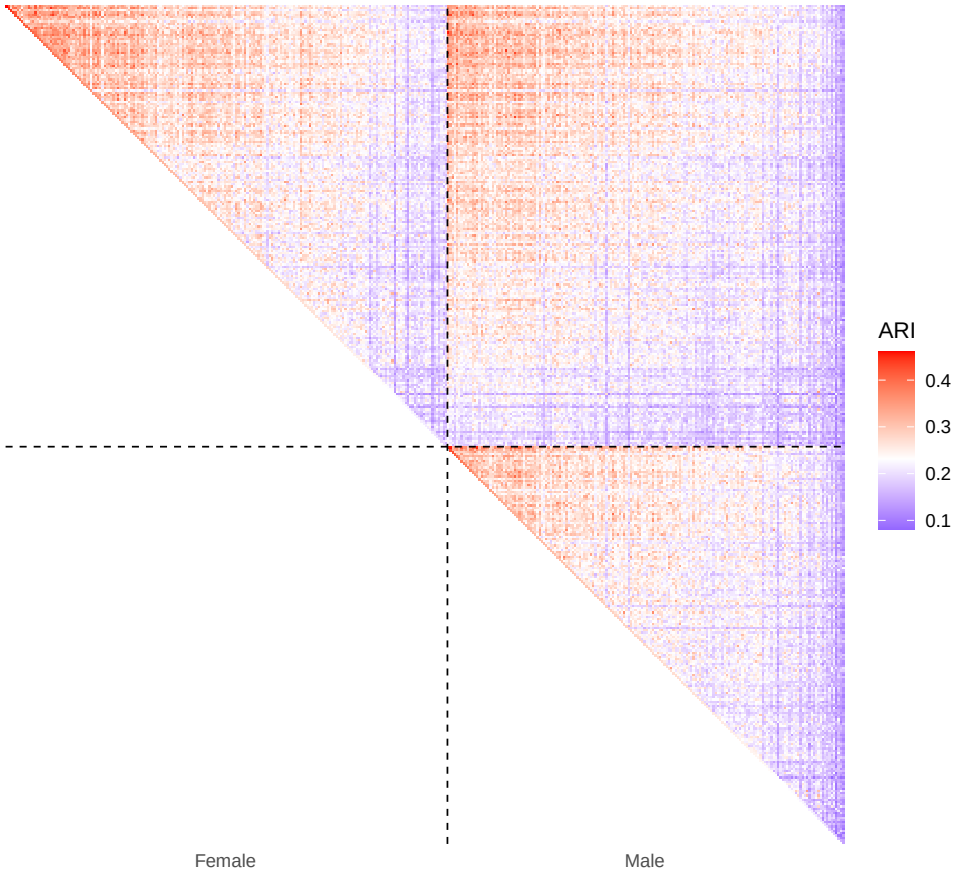


Figure 4: Heat map of ARIs for the community partitions of 376 subject-level brain networks. Each row and column corresponds to a subject. Off-diagonal entries show pairwise ARIs between subjects, while diagonal entries show the ARI between each subject-specific community assignment and CAB-NP. Cells are colored on a diverging blue–white–red scale, with blue and red denoting ARI values below and above the median pairwise ARI, respectively, and white corresponding to the median. Dashed lines separate female and male subjects.

Table 2: The demographic characteristics of subjects in the HCP dataset.

Characteristic	Female (n=198)	Male (n=178)
Age, years (mean ± SD)	29.4 ± 3.6	27.9 ± 3.7
BMI, mean ± SD	25.76 ± 5.01	26.64 ± 4.27
Race, n (%)		
White	146 (73.7)	140 (78.7)
Black or African Am.	33 (16.7)	21 (11.8)
Asian/Nat. Hawaiian/Othr Pacific Is.	10 (5.1)	13 (7.3)
More than one	3 (1.5)	2 (1.1)
Unknown or Not Reported	6 (3.0)	2 (1.1)
Ethnicity, n (%)		
Not Hispanic/Latino	180 (90.9)	157 (88.2)
Hispanic/Latino	15 (7.6)	19 (10.7)
Unknown or Not Reported	3 (1.5)	2 (1.1)

structure of the true underlying network in the presence of network noise. Simulation studies show improved performance of the proposed method over several existing approaches. Analyses of functional brain networks from the HCP and ADNI datasets show that our method yields interpretable community structures, captures subject-specific variation consistent with population-level functional organization in healthy young adults, and identifies systematic patterns of network reorganization associated with disease severity.

Beyond neuroscience, our method is applicable to a wide range of fields involving noisy networks and node-level covariates. For example, in social networks, edges representing interactions or friendships are usually partially observed or misreported (Butts, 2003; Kossinets, 2006; Newman, 2018), while node covariates may include demographic and behavioral attributes. In biological networks, protein–protein or gene interaction networks are often contaminated by experimental false positives and negatives (Chen et al., 2006; Hart et al., 2006; Sprinzak et al., 2003; Von Mering et al., 2002), while node covariates may include gene expression profiles and functional annotations. Jointly leveraging noisy network information and node-level covariates is important for recovering interpretable latent communities across diverse application domains.

Our work also suggests several extensions. First, extending the framework to weighted noisy networks would allow for the use of additional information to better infer the underlying community structure. Second, it would be useful to consider network noise with dependence and/or unstructured heterogeneity. Finally, using nonparametric Bayesian priors to infer the number of communities and developing alternative algorithms such as variational expectation-maximization could reduce the computational cost.

A Appendix: MCMC Algorithm

In this section, we develop an efficient MCMC algorithm for sampling the parameters in our model. We first introduce a data augmentation technique used to facilitate the sampling. We then present the conditional posterior distributions of the model parameters. Finally, we describe the initialization procedure.

A.1 Data Augmentation

We introduce Pólya-Gamma variables for $\boldsymbol{A}$. For simplicity, we denote

$$\psi_{ij} = B_{z_i z_j}\mathbb{I}(z_i \neq z_j) + \theta_i + \theta_j.$$

We note that, for $1 \leq i < j \leq n$,

$$p(A_{ij} \mid z_i, z_j, B_{z_i z_j}, \theta_i, \theta_j) = \frac{\exp(\psi_{ij})^{A_{ij}}}{1+\exp(\psi_{ij})}. \tag{13}$$

Using Theorem 1 in Polson et al. (2013), we can introduce an edge-specific latent Pólya-Gamma variable w_{ij} ($\mathcal{PG}(1,0)$) and rewrite the right hand side of (13) as:

$$\frac{\exp(\psi_{ij})^{A_{ij}}}{1+\exp(\psi_{ij})} = \frac{1}{2}\exp(\kappa_{ij}\psi_{ij})\int_0^\infty \exp\left(-\frac{1}{2}w_{ij}\psi_{ij}^2\right)p(w_{ij})\,dw_{ij},$$

where $\kappa_{ij} = A_{ij} - 1/2$. This leads to the conditional distribution of A_{ij}

$$p(A_{ij} \mid z_i, z_j, B_{z_i z_j}, \theta_i, \theta_j, w_{ij}) \propto \exp\left(\kappa_{ij}\psi_{ij} - \tfrac{1}{2}w_{ij}\psi_{ij}^2\right).$$

A.2 Conditional Posterior Distributions

We generate samples from the posterior density. Let $\boldsymbol{X} = (\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n)^\top$ denote the matrix of p-dimensional nodal covariates, including R continuous and S categorical variables, and let $\tilde{\boldsymbol{A}}^{(1:M)} = \{\tilde{\boldsymbol{A}}^{(m)} : m = 1, \ldots, M\}$ be the collection of all M noisy adjacency matrices. Let Θ denote all model parameters. The posterior density of Θ is given by,

$$\begin{aligned} p(\Theta|\boldsymbol{X}, \tilde{\boldsymbol{A}}^{(1:M)}) \propto\ & p(\boldsymbol{X} \mid \boldsymbol{z}, \boldsymbol{\mu}, \boldsymbol{\tau}^2, \boldsymbol{\xi})\, p(\boldsymbol{\mu} \mid \boldsymbol{\tau}^2)\, p(\boldsymbol{\tau}^2)\, p(\boldsymbol{\xi})\, p(\boldsymbol{z} \mid \boldsymbol{\pi})\, p(\boldsymbol{\pi}) \\ & \times p(\tilde{\boldsymbol{A}}^{(1:M)} \mid \boldsymbol{A}, \boldsymbol{z}, \boldsymbol{\alpha}, \boldsymbol{\beta})\, p(\boldsymbol{\alpha})\, p(\boldsymbol{\beta})\, p(\boldsymbol{A} \mid \boldsymbol{z}, \boldsymbol{\theta}, \boldsymbol{B}, \boldsymbol{w})\, p(\boldsymbol{\theta})\, p(\boldsymbol{B})\, p(\boldsymbol{w}), \end{aligned}$$

where $\boldsymbol{w} = \{w_{ij} : 1 \leq i < j \leq n\}$, $\boldsymbol{\alpha} = (\alpha_{kl})_{K\times K}$, $\boldsymbol{\beta} = (\beta_{kl})_{K\times K}$; $\boldsymbol{\mu} = (\mu_{kr})_{K\times R}$, $\boldsymbol{\tau}^2 = (\tau_{kr}^2)_{K\times R}$, and $\boldsymbol{\xi} = \{\boldsymbol{\xi}_{ks} : k = 1, \ldots, K, s = 1, \ldots, S\}$.

Next, we show the conditional distribution for each parameter.

(1) The conditional distribution of A_{ij}. We use Gibbs sampling to sample A_{ij} from its conditional distribution, that is

$$A_{ij} \mid \tilde{\boldsymbol{A}}^{(1:M)}, z_i, z_j, B_{z_i z_j}, \theta_i, \theta_j, \alpha_{z_i z_j}, \beta_{z_i z_j} \sim \text{Bern}(p_{ij}^*),$$

where

$$p_{ij}^* = \text{logit}^{-1}\left(\psi_{ij} + \ln\left[\left(\frac{1-\beta_{z_iz_j}}{\alpha_{z_iz_j}}\right)^{\sum_{m=1}^M \mathbb{I}(\tilde{A}_{ij}^{(m)}=1)}\left(\frac{\beta_{z_iz_j}}{1-\alpha_{z_iz_j}}\right)^{\sum_{m=1}^M \mathbb{I}(\tilde{A}_{ij}^{(m)}=0)}\right]\right).$$

(2) The conditional distribution of w_{ij}. We use Gibbs sampling to sample w_{ij} from its conditional distribution, that is

$$w_{ij} \mid A_{ij}, z_i, z_j, B_{z_iz_j}, \theta_i, \theta_j \sim \mathcal{PG}(1, \psi_{ij}).$$

(3) The conditional distributions of $\boldsymbol{B}$ and $\boldsymbol{\theta}$. We define $\boldsymbol{\delta} = (\boldsymbol{b}^\top, \boldsymbol{\theta}^\top)^\top$, where $\boldsymbol{b} = (\{B_{kl} : 1 \le k < l \le K\})^\top$, with the pairs (k, l) ordered column-wise over the upper triangle, i.e., $(1,2), (1,3), (2,3), \ldots, (K-1, K)$. Let $\boldsymbol{Y}$ denote the design matrix associated with Model (2), of dimension $\binom{n}{2} \times \left[\binom{K}{2} + n\right]$. Its rows are indexed by the node pairs (i, j) with $i < j$, ordered column-wise over the upper triangle, and its columns correspond to the entries of $\boldsymbol{\delta}$. Let $\boldsymbol{y}_{ij}^\top$ denote the row of $\boldsymbol{Y}$ corresponding to the node pair (i, j). Then, we have

$$\boldsymbol{y}_{ij}^\top \boldsymbol{\delta} = B_{z_iz_j}\mathbb{I}(z_i \ne z_j) + \theta_i + \theta_j = \psi_{ij}.$$

As an example, consider $n = 4$ nodes with $K = 3$ communities and $\boldsymbol{z} = (1, 1, 2, 3)$. In this case $\boldsymbol{\delta} = (B_{12}, B_{13}, B_{23}, \theta_1, \theta_2, \theta_3, \theta_4)^\top$, and the design matrix $\boldsymbol{Y}$ is

$$\boldsymbol{Y} = \begin{array}{c} \begin{array}{ccc|cccc} B_{12} & B_{13} & B_{23} & \theta_1 & \theta_2 & \theta_3 & \theta_4 \end{array} \\ \left[\begin{array}{ccc|cccc} 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 & 1 \end{array}\right] \end{array} \begin{array}{c} \\ (1,2) \\ (1,3) \\ (2,3) \\ (1,4) \\ (2,4) \\ (3,4) \end{array}.$$

The conditional distribution of $\boldsymbol{\delta}$ is

$$\boldsymbol{\delta} \mid \boldsymbol{A}, \boldsymbol{w}, \boldsymbol{z} \sim \mathcal{N}(\boldsymbol{\eta}, \boldsymbol{V})\,\mathbb{I}(\boldsymbol{b} \le \boldsymbol{0}),$$

where $\boldsymbol{V} = \left(\boldsymbol{Y}^\top \boldsymbol{\Omega} \boldsymbol{Y} + \frac{1}{\sigma^2}\boldsymbol{I}_{n+K(K-1)/2}\right)^{-1}$ and $\boldsymbol{\eta} = \boldsymbol{V}\boldsymbol{Y}^\top \boldsymbol{\kappa}$, with $\boldsymbol{\kappa} = \{\kappa_{ij} : 1 \le i < j \le n\}^\top$ and $\boldsymbol{\Omega} = \text{diag}(\{w_{ij} : 1 \le i < j \le n\})$, ordered consistently with the rows of $\boldsymbol{Y}$.

We partition the parameters, mean, and covariance matrix into blocks corresponding to $\boldsymbol{b}$ and $\boldsymbol{\theta}$:

$$\boldsymbol{\delta} = \begin{pmatrix} \boldsymbol{b} \\ \boldsymbol{\theta} \end{pmatrix}, \quad \boldsymbol{\eta} = \begin{pmatrix} \boldsymbol{\eta}_b \\ \boldsymbol{\eta}_\theta \end{pmatrix}, \quad \boldsymbol{V} = \begin{pmatrix} \boldsymbol{V}_{bb} & \boldsymbol{V}_{b\theta} \\ \boldsymbol{V}_{\theta b} & \boldsymbol{V}_{\theta\theta} \end{pmatrix}.$$

We sample $\boldsymbol{\delta}$ by first sampling $\boldsymbol{\theta}$ marginally from its conditional distribution,

$$\boldsymbol{\theta} \mid \boldsymbol{A}, \boldsymbol{w}, \boldsymbol{z} \sim \mathcal{N}(\boldsymbol{\eta}_\theta, \boldsymbol{V}_{\theta\theta}),$$

followed by sampling $\boldsymbol{b}$ from a truncated normal distribution

$$\boldsymbol{b} \mid \boldsymbol{\theta}, \boldsymbol{A}, \boldsymbol{w}, \boldsymbol{z} \sim \mathcal{N}\left(\boldsymbol{\eta}_b + \boldsymbol{V}_{b\theta}\boldsymbol{V}_{\theta\theta}^{-1}(\boldsymbol{\theta} - \boldsymbol{\eta}_\theta), \boldsymbol{V}_{bb} - \boldsymbol{V}_{b\theta}\boldsymbol{V}_{\theta\theta}^{-1}\boldsymbol{V}_{\theta b}\right)\mathbb{I}(\boldsymbol{b} \le \boldsymbol{0}),$$

and then reconstructing the matrix $\boldsymbol{B}$ from $\boldsymbol{b}$.

(4) The conditional distributions of α_{kl} and β_{kl}. We use Gibbs sampling to sample α_{kl} and β_{kl} from their conditional distributions, which are

$$\alpha_{kl} \mid \tilde{\boldsymbol{A}}^{(1:M)}, \boldsymbol{A}, \boldsymbol{z} \sim \text{Beta}\Bigg(a_\alpha + \sum_{1\le i<j\le n} \mathbb{I}\big(\{z_i, z_j\} = \{k, l\}\big)\mathbb{I}(A_{ij} = 0) \sum_{m=1}^{M} \mathbb{I}(\tilde{A}_{ij}^{(m)} = 1),$$
$$b_\alpha + \sum_{1\le i<j\le n} \mathbb{I}\big(\{z_i, z_j\} = \{k, l\}\big)\mathbb{I}(A_{ij} = 0) \sum_{m=1}^{M} \mathbb{I}(\tilde{A}_{ij}^{(m)} = 0)\Bigg),$$
$$\beta_{kl} \mid \tilde{\boldsymbol{A}}^{(1:M)}, \boldsymbol{A}, \boldsymbol{z} \sim \text{Beta}\Bigg(a_\beta + \sum_{1\le i<j\le n} \mathbb{I}\big(\{z_i, z_j\} = \{k, l\}\big)\mathbb{I}(A_{ij} = 1) \sum_{m=1}^{M} \mathbb{I}(\tilde{A}_{ij}^{(m)} = 0),$$
$$b_\beta + \sum_{1\le i<j\le n} \mathbb{I}\big(\{z_i, z_j\} = \{k, l\}\big)\mathbb{I}(A_{ij} = 1) \sum_{m=1}^{M} \mathbb{I}(\tilde{A}_{ij}^{(m)} = 1)\Bigg).$$

(5) The conditional distributions of continuous covariate parameters μ_{kr} and τ_{kr}^2. We use Gibbs sampling to sample μ_{kr} and τ_{kr}^2 from their conditional distributions, which are

$$\mu_{kr} \mid \{x_{ir}^{\text{cont}} : z_i = k\}, \tau_{kr}^2 \sim \mathcal{N}\left(\frac{\mu_{0r}/\iota_{0r} + \sum_{i=1}^{n} \mathbb{I}(z_i = k)\, x_{ir}^{\text{cont}}}{1/\iota_{0r} + N_k}, \quad \frac{\tau_{kr}^2}{1/\iota_{0r} + N_k}\right),$$
$$\tau_{kr}^2 \mid \{x_{ir}^{\text{cont}} : z_i = k\}, \mu_{kr} \sim \mathcal{IG}\Bigg(a_{\tau r} + \frac{1 + N_k}{2},$$
$$b_{\tau r} + \frac{(\mu_{kr} - \mu_{0r})^2}{2\iota_{0r}} + \frac{1}{2}\sum_{i=1}^{n} \mathbb{I}(z_i = k)\left(x_{ir}^{\text{cont}} - \mu_{kr}\right)^2\Bigg).$$

(6) The conditional distribution of categorical covariate parameters $\boldsymbol{\xi}_{ks}$. We use Gibbs sampling to sample $\boldsymbol{\xi}_{ks}$ from its conditional distribution, that is

$$\boldsymbol{\xi}_{ks} \mid \{x_{is}^{\text{cat}} : z_i = k\} \sim \text{Dir}\left(\gamma_s + \sum_{z_i=k} \mathbb{I}(x_{is}^{\text{cat}} = 1),\ \ldots,\ \gamma_s + \sum_{z_i=k} \mathbb{I}(x_{is}^{\text{cat}} = C_s)\right).$$

(7) The conditional distributions of community mixing proportions $\boldsymbol{\pi}$. We use Gibbs sampling to sample $\boldsymbol{\pi}$ from its conditional distribution, that is

$$\boldsymbol{\pi} \mid \boldsymbol{z}, d \sim \text{Dir}\left(\frac{d}{K} + N_1, \ldots, \frac{d}{K} + N_K\right).$$

(8) The conditional distribution of community label $\boldsymbol{z}$. We use Gibbs sampling to sample $\boldsymbol{z}$ from its conditional distribution, that is

$$p(z_i = k \mid \tilde{\boldsymbol{A}}^{(1:M)}, \boldsymbol{A}, \boldsymbol{w}, \boldsymbol{X}, \boldsymbol{z}_{-i}, \boldsymbol{\pi}, \boldsymbol{\theta}, \boldsymbol{B}, \boldsymbol{\mu}, \boldsymbol{\tau}^2, \boldsymbol{\xi}, \boldsymbol{\alpha}, \boldsymbol{\beta})$$
$$\propto \pi_k \times \prod_{j\neq i}\prod_{m=1}^{M} \left[(1 - \beta_{kz_j})^{\mathbb{I}(\tilde{A}_{ij}^{(m)}=1)} \beta_{kz_j}^{\mathbb{I}(\tilde{A}_{ij}^{(m)}=0)}\right]^{A_{ij}} \left[\alpha_{kz_j}^{\mathbb{I}(\tilde{A}_{ij}^{(m)}=1)} (1 - \alpha_{kz_j})^{\mathbb{I}(\tilde{A}_{ij}^{(m)}=0)}\right]^{1-A_{ij}}$$

$$\times \exp\left(\sum_{j\neq i}\left[\kappa_{ij}\psi_{ij}^{(k)} - \frac{1}{2}w_{ij}(\psi_{ij}^{(k)})^2\right]\right)$$
$$\times \prod_{r=1}^{R}\frac{1}{\sqrt{\tau_{kr}^2}}\exp\left(-\frac{(x_{ir}^{\text{cont}}-\mu_{kr})^2}{2\tau_{kr}^2}\right)$$
$$\times \prod_{s=1}^{S}\xi_{ksx_{is}^{\text{cat}}},$$

where $\psi_{ij}^{(k)} = B_{kz_j}\mathbb{I}(k \neq z_j) + \theta_i + \theta_j$, $\boldsymbol{z}_{-i} = (z_1, \ldots, z_{i-1}, z_{i+1}, \ldots, z_n)$ denotes the community assignments of all nodes except node i.

A.3 Initialization

Motivated by Le et al. (2018), we propose the following initialization strategy to facilitate faster MCMC convergence.

(1) Initialization of $\boldsymbol{A}$: We set $A_{ij}^{\text{init}} = \mathbb{I}(\sum_{m=1}^{M}\tilde{A}_{ij}^{(m)} \geq M/2)$.

(2) Initialization of $\boldsymbol{z}$: We initialize the community assignments $\boldsymbol{z}^{\text{init}}$ using spectral clustering (Ng et al., 2002) applied to the initial adjacency matrix $\boldsymbol{A}^{\text{init}}$.

(3) Initialization of $\boldsymbol{\theta}$ *and* $\boldsymbol{B}$: We first calculate the edge density between community k and l based on $\boldsymbol{A}^{\text{init}}$ and $\boldsymbol{z}^{\text{init}}$, which is given by

$$o_{kl}^{\text{init}} = \frac{\sum_{i<j}\mathbb{I}\big(\{z_i^{\text{init}}, z_j^{\text{init}}\} = \{k,l\}\big)A_{ij}^{\text{init}}}{\sum_{i<j}\mathbb{I}\big(\{z_i^{\text{init}}, z_j^{\text{init}}\} = \{k,l\}\big)}.$$

We initialize $\theta_i^{\text{init}} = \frac{1}{2}\operatorname{logit}\big(o_{kk}^{\text{init}}\big)$ for all $i \in \{i : z_i^{\text{init}} = k\}$, and set $B_{kl}^{\text{init}} = \operatorname{logit}\big(o_{kl}^{\text{init}}\big) - \frac{1}{2}\operatorname{logit}\big(o_{kk}^{\text{init}}\big) - \frac{1}{2}\operatorname{logit}\big(o_{ll}^{\text{init}}\big)$.

(4) Initialization of α_{kl} *and* β_{kl}: We initialize α_{kl} and β_{kl} based on the false positive and false negative rates of noisy networks relative to $\boldsymbol{A}^{\text{init}}$. Specifically, we set

$$\alpha_{kl}^{\text{init}} = \frac{\sum_{i<j}\mathbb{I}\big(\{z_i^{\text{init}}, z_j^{\text{init}}\} = \{k,l\}\big)\left(\sum_{m=1}^{M}\tilde{A}_{ij}^{(m)}\right)\mathbb{I}\big(A_{ij}^{\text{init}} = 0\big)}{M\sum_{i<j}\mathbb{I}\big(\{z_i^{\text{init}}, z_j^{\text{init}}\} = \{k,l\}\big)\mathbb{I}\big(A_{ij}^{\text{init}} = 0\big)},$$
$$\beta_{kl}^{\text{init}} = \frac{\sum_{i<j}\mathbb{I}\big(\{z_i^{\text{init}}, z_j^{\text{init}}\} = \{k,l\}\big)\left(M - \sum_{m=1}^{M}\tilde{A}_{ij}^{(m)}\right)\mathbb{I}\big(A_{ij}^{\text{init}} = 1\big)}{M\sum_{i<j}\mathbb{I}\big(\{z_i^{\text{init}}, z_j^{\text{init}}\} = \{k,l\}\big)\mathbb{I}\big(A_{ij}^{\text{init}} = 1\big)}.$$

(5) Initialization of covariate parameters: We initialize the parameters associated with x_{ir}^{cont} using the sample means and variances within each community defined by $\boldsymbol{z}^{\text{init}}$, which are

$$\mu_{kr}^{\text{init}} = \frac{\sum_{i=1}^{n}\mathbb{I}\big(z_i^{\text{init}} = k\big)\,x_{ir}^{\text{cont}}}{\sum_{i=1}^{n}\mathbb{I}\big(z_i^{\text{init}} = k\big)},\quad \tau_{kr}^{2,\text{init}} = \frac{\sum_{i=1}^{n}\mathbb{I}\big(z_i^{\text{init}} = k\big)\,\big(x_{ir}^{\text{cont}} - \mu_{kr}^{\text{init}}\big)^2}{\sum_{i=1}^{n}\mathbb{I}\big(z_i^{\text{init}} = k\big)},\quad k = 1, \ldots, K.$$

Similarly, we initialize the parameters for categorical covariates as

$$\xi_{ksc}^{\text{init}} = \frac{\sum_{i=1}^{n}\mathbb{I}\big(z_i^{\text{init}} = k,\ x_{is}^{\text{cat}} = c\big)}{\sum_{i=1}^{n}\mathbb{I}\big(z_i^{\text{init}} = k\big)},\quad c = 1, \ldots, C_s.$$

Acknowledgements

This work was supported in part by NIH Grant U01 AG068057. This work was also supported by NIH Grants F31 AG091902 and R01 AG071174. The content is solely the responsibility of the authors and does not necessarily represent the official views of the National Institutes of Health.

The complete ADNI Acknowledgment is available at https://adni.loni.usc.edu/wp-content/uploads/how_to_apply/ADNI_Acknowledgement_List.pdf.

HCP data were provided by the Human Connectome Project, WU-Minn Consortium (Principal Investigators: David Van Essen and Kamil Ugurbil; 1U54MH091657) funded by the 16 NIH Institutes and Centers that support the NIH Blueprint for Neuroscience Research; and by the McDonnell Center for Systems Neuroscience at Washington University.

Data Availability

The data that support the findings in this paper are available in the Alzheimer's Disease Neuroimaging Initiative database at https://adni.loni.usc.edu and the Human Connectome Project database at http://www.humanconnectomeproject.org/data/.

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